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II Semester M.C.A. Examination, Oct./Nov. 2020
(CBCS Scheme)
COMPUTER SCIENCE
MCA201T : Data Structures

Time : 3 Hours Max. Marks : 70

Instruction : *Answer* ***any five*** *questions from Section – A and* ***any four*** *from Section – B.*

SECTION – A

Answer **any five** questions. **Each** question carries **six** marks : **(5×6=30)**

1. What is data structure ? Discuss the operations performed on data structures.
2. Explain the different types of asymptotic notations with examples.
3. Write and trace the algorithm for insertion sort for the given numbers : 15, 12, 7, 20, 8, 55.
4. Write a program to implement operations on singly linked lists.
5. Write an algorithm for postfix evaluation.
6. Explain the array implementation of ordinary queue for insertion and deletion.
7. Write an algorithm for DFS and explain with suitable example.
8. What are the different binary search trees available ? Explain.

SECTION – B

Answer **any four** questions. **Each** question carries **10** marks : **(4×10=40)**

9. a) Write a program to perform binary search. 5
   b) Explain Knuth Morris Pratt pattern matching algorithm. 5
10. a) Explain tower of Hanoi for n = 3 discs. 5
    b) Write a program for merge sort technique. 5

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11. a) Write linked list implementation for stack operations. 6

    b) Explain the various types of queues with example. 4

12. Write and trace algorithms for various types of tree traversals. 10

13. a) Define graph. Explain BFS graph traversal. 4

    b) Write a program to delete a node from binary search tree. 6

14. Write short notes on : (2×5=10)

    i) Quick sort.

    ii) Boyer Moore string pattern matching.

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I Semester M.C.A. (2 Years Course) Examination, August/September 2021
(CBCS Scheme)
(2020 – 2021 and Onwards)
Computer Science
1MCA6 : DATA STRUCTURES

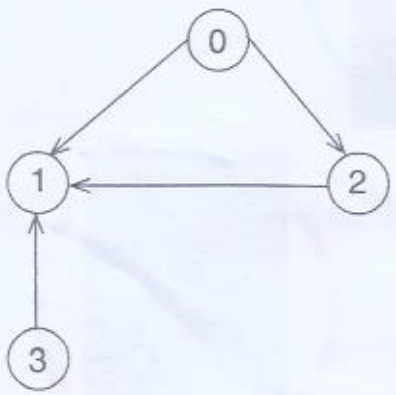
Time : 3 Hours Max. Marks : 70

Instructions : 1) *Answer **all** the Sections.*
2) *Answer **any five** from Section – **A**, Answer **any four** from Section – **B**.*

SECTION – A

Answer **any five** of the following. **Each** question carries **six** marks. **(5×6=30)**

1. Define data structure and explain the operations performed on the linear data structure. 6
2. Explain abstract data type. What is the importance of abstract data type in data structure ? 6
3. What are the ways in which a two dimension array is stored ? Write any two limitations of array. 6
4. Given the following sparse matrix,
$$\begin{bmatrix} 0 & -1 & 0 & 0 & 2 \\ 3 & 0 & 4 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 2 & 0 & 1 & 0 \end{bmatrix}$$
. Design an algorithm to search an item in sparse matrix. 6
5. What is weighted graph ? Write the adjacency matrix for the following graph. 6



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6. Convert the following infix expression to postfix using stack
   A * B – (f + D) + E. 6

7. Explain different collision resolution strategies for hashing. 6

8. Explain Quick sort algorithm in detail with an example. 6

SECTION – B

Answer **any four** of the following. **Each** question carries **ten** marks. **(4×10=40)**

9. a) Explain string as ADT. 6
   b) Write an algorithm to delete a substring from a given string. 4

10. a) Write an algorithm to traverse in a single linked list. 6
    b) Write an algorithm to insert a node at the beginning of a singly linked list. 4

11. a) Write an algorithm for push and POP operations on stack. 6
    b) Write an algorithm to find GCD of two numbers using recursion. 4

12. a) Explain directed and undirected graph with an example. 6
    b) Discuss the importance of graph and tree in data structure. 4

13. a) Explain depth first search algorithm to traverse a graph. 6
    b) Explain linear search with its merits and demerits. 4

14. a) Write bubble sort algorithm. Sort the following using bubble sort.
    {5, 1, 6, 2, 4, 3}. 6
    b) Explain Binary search algorithm. 4

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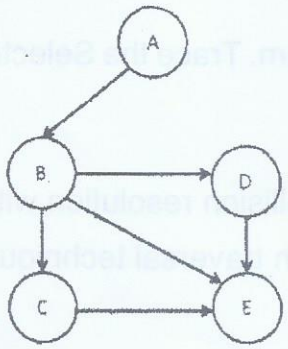
I Semester M.C.A. (2 Years Course) Examination, July 2022
(CBCS – 2020 – 21 and Onwards)
COMPUTER SCIENCE
1MCA6 : Data Structures

Time : 3 Hours Max. Marks : 70

SECTION – A

Answer **any five** of the following. **Each** question carries **six** marks. **(5×6=30)**

1. Define Data structure. Explain the operations performed on data structures with example.
2. What do you mean by Asymptotic behavior of a function ? Explain the different types of Asymptotic notations.
3. Write and trace the Insertion sort algorithm for the given set of elements. {10, 8, 22, 5, 30, 7, 19}.
4. Convert the following infix expression to postfix expression using stack A – (B/C)*D + (E^F).
5. What is stack ? Explain Tower of Hanoi problem for the number of discs=3.
6. Write an algorithm for PUSH and POP operations on stack.
7. Write an algorithm to insert an element in the beginning of the linked list.
8. What is Adjacency matrix of graph ? Write the adjacency matrix for the given graph.



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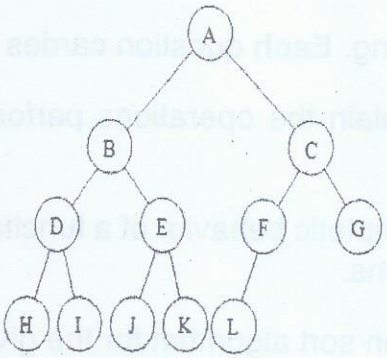
SECTION – B

Answer **any four** of the following. **Each** question carries **ten** marks. **(4×10=40)**

9. a) Define a string. Write a program to concatenate two strings without using built-in function. **6**

   b) What is an array ? Write an algorithm to insert an element in to the array. **4**

10. a) Define traversing. Write the Preorder, Inorder and Postorder traversing for the given tree. **6**



   b) Define recursion. Explain a recursive algorithm to find GCD of two numbers. **4**

11. a) Write an algorithm to create a Binary tree. Trace the Binary tree insertion algorithm for the given set of elements and show the tree diagram for all the steps. **7**

   {7, 2, 5, 1, 9, 3, 4, 8}.

   b) Write an algorithm for Binary search. **3**

12. a) Write an algorithm to insert and delete the elements from Circular Queue. **6**

   b) What is 2-D array ? Write a program to check whether the matrix is sparse matrix or not. **4**

13. a) Write an algorithm for insertion and deletion operations for a circular linked list. **5**

   b) Write Selection sort algorithm. Trace the Selection sort method for the given set of values. **5**

   {10, 45, 12, 23, 6, 33}.

14. a) Explain the technique of collision resolution with open addressing. **5**

   b) Write an algorithm for graph traversal techniques. **5**

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I Semester M.C.A. (Two Years Course) Examination, July 2023
(CBCS) (2020 – 2021 and Onwards)
COMPUTER SCIENCE
1MCA6 : Data Structures

Time : 3 Hours Max. Marks : 70

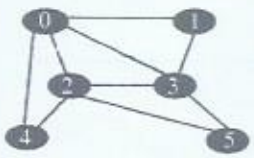
Instructions : *1) Answer **any five** questions from Section – **A**.*
*2) Answer **any four** questions from Section – **B**.*

SECTION – A

I. Answer **any five** of the following. **Each** question carries **6** marks. **(5×6=30)**

1) Define data structures. Explain the classification of data structures.
2) What are Asymptotic Notations ? Explain with an example.
3) Define an Array ADT. Write a function to search an element in an array.
4) What is a Polish Expression ? Evaluate the following postfix expression using a stack table.

   2 3 * 4 5 + *
5) Define the following :
   i) Binary tree
   ii) Binary search tree
   iii) Complete binary tree.
6) Define a weighted graph. Write the adjacency matrix for the following graph.



7) Write an algorithm to perform quick sort for contiguous list with example.
8) Explain the collision resolution using open addressing with an example.

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SECTION – B

II. Answer **any four** of the following. **Each** question carries **10** marks. **(4×10=40)**

9) a) Write an algorithm to perform Bubble Sort and illustrate time complexity of an algorithm. **6**

b) Write a function to perform concatenation of two strings. **4**

10) a) Explain the recursive approach to solve the problem of Tower of Hanoi with three discs. **5**

b) Define stack. Explain the PUSH and POP operations of stack with an example. **5**

11) a) Write an algorithm to convert infix expression to postfix expression. **5**

b) Define a Queue. Explain the insert and delete operation in circular queue. **5**

12) a) Write a function for Insertion at End and Deletion at End in Singly Linked List. **6**

b) Write an algorithm to perform insert operation in doubly linked list with an example. **4**

13) Explain the different types of tree traversal techniques with its algorithm and example. **10**

14) a) Sort the given list of elements using Merge Sort. Discuss with algorithmic steps.

{14, 43, 27, 86, 35, 19, 54, 78, 109, 61} **7**

b) Write a note on Hashing. **3**

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I Semester M.C.A. (Two Years Course) Examination, June/July 2024
(CBCS Scheme)
(2020 – 21 and Onwards)
COMPUTER SCIENCE
Paper – 1MCA6 : Data Structures

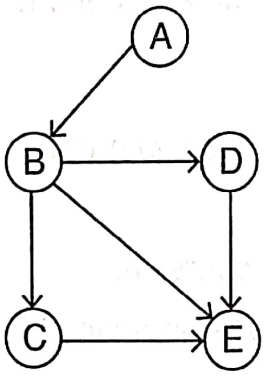
Time : 3 Hours Max. Marks : 70

Instructions *: 1) Answer* ***any five*** *questions from Part –* ***A.***
2) Answer ***any four*** *questions from Part –* ***B.***

PART – A

Answer **any five** of the following. **Each** question carries **six** marks. **(5×6=30)**

1. Define data structures. Explain the classification of data structures.
2. Explain string ADT. What are the operations performed in string (any 3) ?
3. What are the Asymptotic notations ? Explain with example.
4. Convert the infix expression to postfix expression ((A + B) – C*(D/E)) + F.
5. What is a shell sort ? Trace it on the following list of numbers.
   3, –1, 10, 5, 2.
6. What is a graph ? How to represent a graph ? Use the following graph as an example.



7. Write an algorithm to perform for quick sort on the given list.
   10, 80, 30, 90, 40, 50, 70.
8. What is stack ? Explain its operations.

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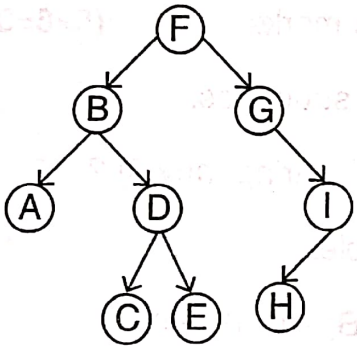
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PART – B

Answer **any 4** questions. **Each** question carries **10** marks. **(4×10=40)**

9. What is a linked list ? Explain the following operations on the linked list.
   a) Insertion at the beg. of list
   b) Insertion at the end of list
   c) Insertion at the middle of list
   d) Searching in a list
   e) Deleting the first node in the list.

10. What is tree traversal ? Write the preorder, postorder and inorder traversal for the following tree.



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11. a) Write selection sort algorithm and trace it for the following list. **5**
    {10, 45, 12, 23, 06, 33}

    b) Write an algorithm for insertion and deletion operations for a circular linked list. **5**

12. a) What is binary search ? Write the algorithm for binary search and trace it for the following list and search for the element 10. **5**
    2 5 7 9 10 20

    b) What is hashing ? Explain the technique of collision resolution with open addressing. **5**

13. a) Define recursion. Explain towers of Hanoi algorithm using recursion. **5**

    b) Write a program to perform insertion and deletion operations on a simple queue. **5**

14. Write notes on :
    a) Garbage collection **5**
    b) AVL Trees. **5**

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I Semester M.C.A. (Two Years Course) Examination, May/June 2025
(CBCS) (2020 – 21 and Onwards)
COMPUTER SCIENCE
1MCA6 : Data Structures

Time : 3 Hours Max. Marks : 70

Instruction *: Answer* ***all*** *the Sections.*

SECTION – A

Answer **any five** of the following. **Each** question carries **six** marks. **(5×6=30)**

1. Explain different types of data structures.
2. Explain the significance of time complexity and space complexity of an algorithm.
3. What is a Sparse Matrix ? Write a C program to check whether a matrix is a triangular Sparse Matrix or not.
4. Explain different type of linked lists.
5. Briefly explain height-balancing in AVL trees.
6. What is topological sorting ? Elaborate.
7. What is divide and conquer method ? Explain.
8. Write a program to sort n numbers using merge-sort.

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SECTION – B

Answer **any four** of the following. **Each** question carries **10** marks. **(4×10=40)**

9. What is an asymptotic notation ? Explain different types of asymptotic notations.
10. a) Convert the following infix notation expression to postfix notations
    $(x + y)/(p - q) - f^*e$ **6**
    b) Explain underflow and overflow conditions with respect to stack and queues. **4**
11. Define Binary Search Tree. Construct binary search tree for the given data. Write the outcome of three tree traversal on the constructed BST.
12. Write functions for the following operations using singly linked list.
    i) Insertion at beginning
    ii) Deletion of first node
    iii) Search a given key in a list.
13. Write selection sort algorithm and trace it for the given data.
    150, 30, 60, 95, 7, 80, 50, 100.
14. Define Hashing. Explain hashing techniques and techniques for collision resolution.
